

Text-to-Motion Generation with Discrete Representations and Large Language Models

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Abstract—Based on Vector Quantized Variational AutoEncoder (VQ-VAE) and Transformers, we investigate a simple yet effective conditional generative framework for text-to-motion generation. First, we find that, with a few training recipes (EMA and Code Reset), a CNN-based VQ-VAE can learn high-quality discrete motion representations, which largely facilitate motion generation. Second, to better understand motion descriptions, we propose T2M-GPT+ that uses Large Language Models (LLMs) to extract text features. Further, we propose T2M-GIT+, which employs a non-autoregressive method to generate discrete motion representations in parallel, and is therefore more efficient than T2M-GPT+ while achieving comparable results. Comprehensive experiments show that our method is superior to existing methods, including competitive diffusion methods. For example, we achieve a Top-1 R-precision of 0.53 and an FID of 0.10 on the HumanML3D dataset, substantially outperforming MotionDiffuse, which obtains a Top-1 R-Precision of 0.49 and an FID of 0.63. Additionally, we conduct analyses and find that, even for the largest dataset, *i.e.*, HumanML3D, it may still limit the performance of the proposed approach. This suggests that a larger dataset could bring additional improvement to our approach. Our implementation is available on the project page: <https://mael-zys.github.io/T2M-GPT/>.

Index Terms—Text-to-Motion Generation, Vector Quantized-Variational AutoEncoder, Generative Model, Large Language Models.

I. INTRODUCTION

GENERATING motion from textual descriptions has gained significant attention due to its potential applications in the game industry, film-making, and animating robots. For instance, in the game industry, accessing new motions typically relies on motion capture, which often requires professional actors and is therefore highly costly. Automatically generating human motion from textual descriptions could save time and be more economical.

Because motion and text are from different modalities, generating motion from natural language is challenging. The designed model is expected to establish an accurate and robust mapping from the language space to the motion space. To address this, several studies propose learning a unified

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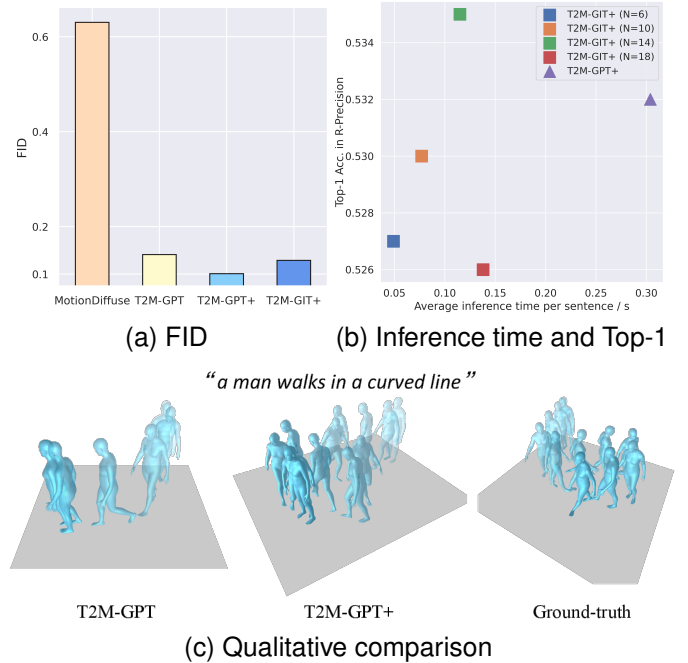


Fig. 1. Comparisons on HumanML3D [7]. T2M-GPT+ and T2M-GIT+ outperform the previous version (T2M-GPT [9]) and the state-of-the-art, MotionDiffuse [10], on (a) FID ↓, (b) average inference time per sentence in seconds ↓ and Top-1 accuracy in R-Precision ↑, where N is the number of inference steps. (c) Our approach can generate more complex motions that closely align with the ground truth than T2M-GPT.

embedding for text and motion via autoencoders [1]–[3] and Variational Autoencoders (VAEs) [4], [5].

For example, MotionClip [3] focuses on aligning motion and CLIP [6] spaces, while ACTOR [4] and TEMOS [5] employ transformer-based VAEs to generate motions from action labels and texts, respectively. However, these approaches struggle with complex descriptions. Subsequent work by Guo *et al.* [7], [8] aimed to address this limitation by generating motions from longer and more detailed texts. However, these methods sometimes fail to maintain text-motion consistency, and their three-stage pipelines introduce complexity to the training process.

Recently, diffusion-based models [11] have demonstrated remarkable performance on image generation [12], [13] and motion generation [10], [14]–[18]. Unlike image generation, which requires generating images in high resolution (typically 512×512), motion generation is relatively easy in output dimensionality ($22 \text{ joints} \times 196 \text{ frames at most}$). Therefore,

we try to explore classic approaches, such as VQ-VAE [19], for text-driven motion generation.

In this paper, motivated by recent advances in learning discrete representations for generative modeling [19]–[26], we investigate a simple and classic framework for text-to-motion generation based on Vector Quantized Variational Autoencoders (VQ-VAEs) [19] and generative transformers [27]–[29].

Specifically, we propose a two-stage method for text-to-motion generation. In stage 1, a standard 1D convolutional network is used to map motion sequences into discrete code indices. In stage 2, a generative transformer is learned to generate sequences of code indices from textual descriptions. During this process, we find that the naive training of VQ-VAE [19] leads to code collapse. To address this issue, we incorporate two standard techniques: Exponential Moving Average (EMA) and *Code Reset*, and provide a detailed analysis of these quantization strategies. For generative transformers, we investigate auto-regressive generation with Generative Pre-trained Transformer [28] (GPT) and non-autoregressive generation with parallel decoding via Generative Image Transformer [29] (GIT). To the best of our knowledge, we are the first to introduce parallel decoding using GIT for human motion generation.

Different from widely used text encoders (*i.e.*, CLIP [6] and DistilBERT [30]) that are trained via contrastive learning and masked language modeling, Large Language Models (LLMs) are commonly used as auto-regressive generators rather than pre-trained models for text embedding. We propose T2M-GPT+ and demonstrate that, without specific fine-tuning, text embeddings generated by LLMs bring a significant improvement for this task. In T2M-GPT+, a single-step prediction error may lead to cascading inaccuracies in future motion code predictions. To alleviate this problem, we propose simulating single-step errors by corrupting sequences during training. Moreover, the inference process in existing approaches, which relies on diffusion or autoregressive methods, is usually time-consuming. This presents a challenge for real-time motion generation applications. We propose T2M-GIT+ to tackle this challenge. We adopt the Masked Token Modeling [29], [31] strategy during training and generate motion codes in a parallel manner, making it more efficient than the serial T2M-GPT+ method. Specifically, T2M-GIT+ is about 2.64 times faster than T2M-GPT+ while achieving comparable results.

Our approach effectively generates high-quality motion sequences that align with textual descriptions (Figure 1). Empirically, we demonstrate superior performance compared to diffusion-based approaches, *i.e.*, MLD [16], MDM [14], and MotionDiffuse [10] on two widely-used datasets: HumanML3D [7] and KIT-ML [32]. For example, T2M-GPT+ with Llama-13B [33] achieves Top-1 accuracy 0.532 in R-precision and FID 0.099 on HumanML3D, significantly outperforming MotionDiffuse with Top-1 accuracy 0.491 and FID 0.630. As large models require high-quality datasets, we also explore the impact of dataset size for this task. The empirical analysis suggests that the performance of our model can be improved with larger datasets.

In summary, our contributions include:

- We propose a simple yet effective approach for text-

driven human motion generation. The proposed approach achieves state-of-the-art results on two widely used benchmarks: HumanML3D [7] and KIT-ML [32].

- We show that the word-wise text embedding from Large Language Models (LLMs) can significantly improve performance without specific fine-tuning.
- We investigate both auto-regressive (T2M-GPT+) and non-autoregressive (T2M-GIT+) generation. T2M-GIT+ is, to our knowledge, the first approach to decode in parallel via GIT.
- We provide a comprehensive analysis of the impact of quantization strategies and dataset size. Our findings suggest that larger datasets might still hold significant potential for advancing text-to-motion generation.

Visual results and the implementation are available on the project page: <https://mael-zys.github.io/T2M-GPT/>.

II. RELATED WORK

A. VQ-VAE

Vector Quantized Variational Autoencoder (VQ-VAE) is introduced in [19] as a variant of the VAE [34]. The effectiveness of VQ-VAE lies in its ability to learn a high-quality discrete representation from data distribution using an autoencoder architecture. It demonstrates strong performance in generative tasks across diverse domains, such as image synthesis [20], [21], text-to-image generation [22], speech-driven gesture generation [23], and music generation [24], [25]. Despite its success, training VQ-VAE often suffers from code collapse, where only a portion of the codes in the codebook are activated, resulting in low-quality reconstruction. Various strategies have been developed to address this challenge, including exponential moving average (EMA) for updating the codebook [20] and Code Reset to reactivate unused entries [20], *etc.* Inspired by the above works, we employ VQ-VAE based on 1D Convolutional Neural Networks (CNNs) to obtain discrete human motion representations.

B. Human motion synthesis

Human motion synthesis has been a well-established and long-standing research area. Human motion prediction conditioned on past motions is one of the challenging problems in this research field. Early efforts utilize recurrent neural networks (RNNs) [35]–[38] to capture temporal dependencies. Other works propose to model spatial-temporal information via GCN [39], Self-attention [40], [41], and MLP [42]–[44]. Some researchers have explored generating diverse motion sequences using GANs [45], [46] and VAEs [47]–[49]. Another active research topic is motion in-betweening [50]–[54], which aims to generate transitions between past and future poses. For the unconstrained generation task, CSGN [55] builds long-range temporal relationships based on the Gaussian process. Motion generation can also be driven by various modalities, such as action [4], [56], music [57]–[62], and speech [63]. In the graphics domain, significant attention has been placed on animator control. For example, Holden *et al.* [64] used a convolutional autoencoder to reconstruct motion, enabling both synthesis and editing. Phase-Functioned Neural Network [65]

proposes a real-time motion synthesis approach for character control. Starke *et al.* [66] employed an auto-regressive model to capture complex human-scene interaction behaviors. Deepphase [67] transforms human motions into a periodic manifold and proposes a periodic autoencoder to enhance motion synthesis. Drawing inspiration from SinGAN [68], GANimator [69] generates diverse and novel motions by training on a single motion sequence.

C. Text-driven human motion generation

Text-driven human motion generation aims to synthesize 3D human motion from textual descriptions. Early works like Text2Action [70] and Language2Pose [1] utilize RNN-based models and curriculum learning to generate motions from short texts, respectively. Ghosh *et al.* [2] further improved on Language2Pose by learning separate representations for upper and lower body movements. Similar to Language2Pose [1] that learns a joint feature space of motion and text, MotionCLIP [3] leverages CLIP [6] as the text encoder and uses rendered images for additional supervision. This method facilitates out-of-distribution motion generation and latent feature-based motion editing. However, the generated motion lacks both diversity and global translation. To address this, TEMOS [5] enhances the ACTOR [4] architecture (Transformer-based VAE for action-to-motion generation) by incorporating an additional text encoder. TEMOS is trained on the KIT Motion-Language dataset [32] and demonstrates its efficacy, but it struggles to generate high-quality motions from out-of-distribution descriptions. Subsequently, TEACH [71] and SINC [72] further build upon TEMOS for temporal and spatial motion compositions, respectively. Recently, Guo *et al.* [7] proposed a large-scale and diverse text-to-motion dataset, *i.e.*, HumanML3D, and introduced a motion-length prediction model to control the length of generated motion. TM2T [8] shows the benefits of joint training text-to-motion and motion-to-text tasks [73]. AttT2M [74] enhances VQ-VAE with body-part attention and GPT with global-local motion-text attention. MotionGPT [75] demonstrates that finetuning a pre-trained Large Language Model (LLM) with LoRA [76] allows human motion generation from multiple types of control. In comparison to MotionGPT [75], our work uses the LLM to obtain word-wise text embedding without fine-tuning. MMM [77] and MoMask [78], published after the initial submission of this paper, share similar motivations with our approach. In particular, MoMask redesigns the VQ-VAE to learn hierarchical motion representations. Additionally, MDM [14] and Motion-Diffuse [10] are early attempts to apply diffusion models to motion generation, followed by subsequent studies with promising performance [16], [79]–[82]. In this work, we show that the discrete latent space benefits motion generation. We also demonstrate that without sophisticated designs, the classic VQ-VAE framework with some standard training recipes could outperform several state-of-the-art diffusion models.

D. Large language models

Language model pre-training is a crucial task in natural language processing. BERT [31] introduces a bidirectional

transformer-based architecture with a masked language modeling strategy, which has shown great improvements. Autoregressive language model [28], [83]–[86] is another pre-training paradigm, showing significant growth in model size in recent years. GPT-3 [87] increases the model size to 175B parameters, which prompts the development of Large Language Models (LLMs). Subsequently, the emergence of several large language models facilitated the development of the community, such as OPT [88], PaLM [89], Llama [33], Vicuna [90], *etc.* A complete review of large language models is outside the scope of this work. A recent survey can be found in [91]. The works mentioned above train the LLM as an autoregressive-based text generator. Different from these methods, we show that a GPT-like LLM can also be a powerful text encoder. We also demonstrate that a stronger LLM-based text encoder can be beneficial for motion generation.

III. METHOD

The overall framework comprises two modules: Motion VQ-VAE and Transformers (T2M-GPT+ and T2M-GIT+), as illustrated in Figure 2. Motion VQ-VAE maps motion data to discrete code sequences, while the Transformers generate code indices based on textual descriptions. We are able to reconstruct the motion from the code indices through the decoder in Motion VQ-VAE. In Section III-A, we introduce the Motion VQ-VAE. We provide details of T2M-GPT+ and T2M-GIT+ in Section III-B and Section III-C, respectively.

A. Motion VQ-VAE

VQ-VAE [19] is a framework that learns discrete representations for generative models. Our VQ-VAE is shown in Figure 2, which consists of an encoder E , a decoder D , and a learnable codebook $C = \{c_k\}_{k=1}^K$ with K codes, where $c_k \in \mathbb{R}^{d_c}$ and d_c denotes the dimension of codes. Specifically, given a T -frame motion sequence $X = [x_1, x_2, \dots, x_T]$, where $x_t \in \mathbb{R}^d$, $t \in [1, 2, \dots, T]$, and d is the dimension of the motion. We first encode X into latent features $Z = E(X)$, where $Z = [z_1, z_2, \dots, z_{T/l}]$ with $z_i \in \mathbb{R}^{d_c}$ and l is the temporal downsampling rate. Then, each z_i is quantized to its most similar element $\hat{z}_i$ in C , which can be written as:

$$\hat{z}_i = \arg \min_{c_k \in C} \|z_i - c_k\|_2. \quad (1)$$

Optimization goal. To optimize the Motion VQ-VAE, we reconstruct quantized latent representations $\hat{Z}$ back to X_{re} through the decoder $X_{re} = D(\hat{Z})$. The optimization objective $\mathcal{L}_{vq}$ is computed as:

$$\mathcal{L}_{vq} = \mathcal{L}_{re} + \underbrace{\|sg[Z] - \hat{Z}\|_2}_{\mathcal{L}_{embed}} + \beta \underbrace{\|Z - sg[\hat{Z}]\|_2}_{\mathcal{L}_{commit}}, \quad (2)$$

where $\mathcal{L}_{embed}$, $\mathcal{L}_{commit}$, and β denote the embedding loss, the commitment loss, and the weight of the commitment loss, respectively. For the reconstruction loss $\mathcal{L}_{re}$, we incorporate velocity regularization to improve the motion quality:

$$\mathcal{L}_{re} = \mathcal{L}_1^{smooth}(X, X_{re}) + \alpha \mathcal{L}_1^{smooth}(V(X), V(X_{re})), \quad (3)$$

where α is a hyperparameter to balance the two terms, $\mathcal{L}_1^{smooth}$ is the Smooth L1 loss, and $V(X) = [v_1, v_2, \dots, v_{T-1}]$ is

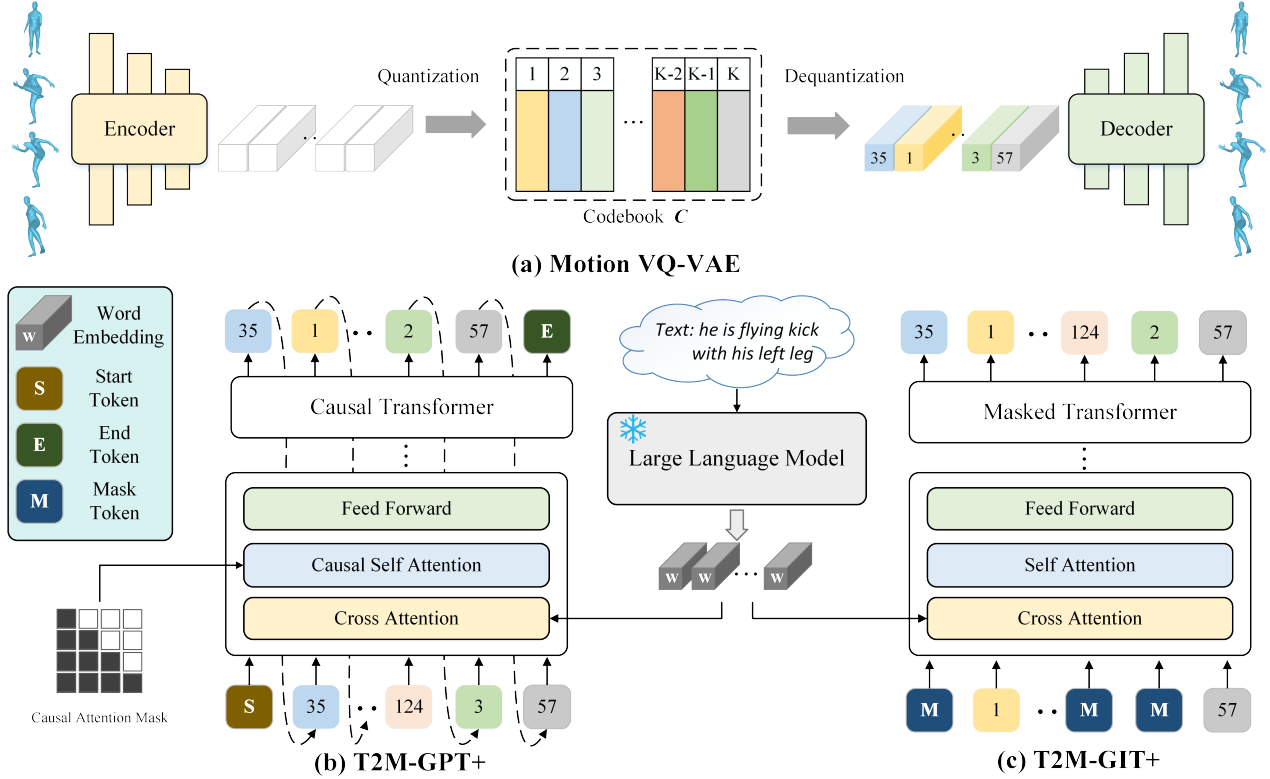


Fig. 2. Overview of our framework for text-to-motion generation. It consists of two modules: Motion VQ-VAE and Transformers (T2M-GPT+ and T2M-GIT+). (a) Motion VQ-VAE learns a mapping between motion data and discrete code sequences. Either (b) T2M-GPT+ or (c) T2M-GIT+ can be used to generate code indices with text features extracted by Large Language Models. Note that T2M-GPT+ and T2M-GIT+ are two separate models and are trained independently.

the velocity of X with $v_i = x_{i+1} - x_i$. We conduct ablative experiments on α , and compare different reconstruction losses, including $\mathcal{L}_1$, $\mathcal{L}_1^{\text{smooth}}$, and $\mathcal{L}_2$ in Section IV-D.

Quantization strategy. We employ two commonly used quantization recipes [92], *i.e.*, exponential moving average (EMA) and codebook reset (*Code Reset*), to address the codebook collapse [19], [92] problem. EMA facilitates smooth codebook updating through $C^i \leftarrow \lambda C^{i-1} + (1 - \lambda)C^i$, where C^i denotes the codebook at iteration i and λ is the moving average constant. Meanwhile, *Code Reset* dynamically reassigns inactive codes based on the input data distribution. The ablation study of these strategies is provided in Section IV-D.

Architecture. The detailed architecture of motion VQ-VAE is shown in Figure 3. Inspired by [9], [21], [62], our model is composed of convolutions, ReLU, and residual blocks. Denote the downsampling rate as $l = 2^L$, where L represents the number of residual blocks in the encoder (or decoder). Temporal downsampling is achieved using convolutions with a stride of 2, while nearest-neighbor interpolation is utilized for upsampling.

B. T2M-GPT+

Given a motion sequence $X = [x_1, x_2, \dots, x_T]$, a learned VQ-VAE is used to map X to a sequence of code indices $S = [s_1, s_2, \dots, s_{T/l}]$, where $s_i \in [1, 2, \dots, K]$ is the index in the codebook. We introduce two special tokens, *i.e.*, [START] and [END], at the beginning and end of the sequence,

indicating the start and stop of the motion, respectively. Note that this design differs from [7], which leverages an extra model to predict motion length. To reconstruct the motion, we feed S into a transformer-based model [27], obtaining the corresponding codebook entries and the discrete representation $\hat{Z} = [\hat{z}_1, \hat{z}_2, \dots, \hat{z}_{T/l}]$, where $\hat{z}_i = c_{s_i}$. The motion X_{re} can be projected back from $\hat{Z}$ via the decoder D . We formulate the task of text-driven motion generation as a next-index prediction problem. Specifically, with a text condition c and the previously generated indices $S_{<i}$, we aim to generate the next index s_i by predicting the probability distribution $p(s_i|c, S_{<i})$. The illustration of T2M-GPT+ is shown in Figure 2 (b).

Optimization goal. To optimize the model, we minimize the negative log-likelihood of $p(S|c)$, where $p(S|c) = \prod_{i=1}^{|S|} p(s_i|c, S_{<i})$. The training objective becomes:

$$\mathcal{L}_{GPT} = \mathbb{E}_{S \sim p(S)} [-\log p(S|c)]. \quad (4)$$

Large language models. Motivated by the great success of Large Language Models [33], [89], [90], we propose to use Llama [33] as the text feature extractor. Specifically, we denote the word-wise text embedding as $e \in \mathbb{R}^{T' \times d_t}$ with text length T' and the dimension of the word embedding d_t . The text embedding can be computed as $e = LLM(c)$. We consider integrating the text embedding into the transformer with the cross-attention operator:

$$CA = \text{Softmax} \left(\frac{Q(m) \cdot K(e)^T}{\sqrt{d_k}} \right) \cdot V(e), \quad (5)$$

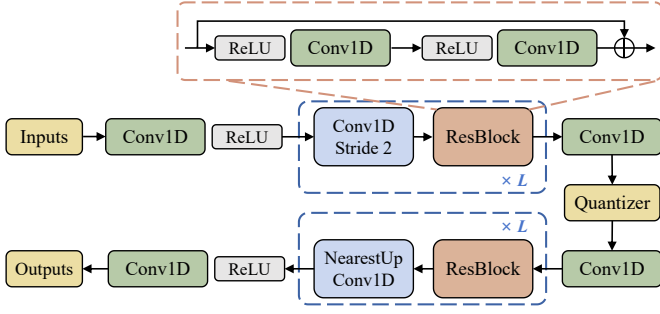


Fig. 3. The architecture of the motion VQ-VAE, which consists of ‘ L ’ residual blocks. Temporal downsampling is achieved using convolutions with a stride of 2, while nearest-neighbor interpolation is used for upsampling.

where m denotes the embedding of motion code indices, $Q(m) \in \mathbb{R}^{\frac{T}{l} \times d_k}$ is the query, $K(e) \in \mathbb{R}^{T' \times d_k}$ and $V(e) \in \mathbb{R}^{T' \times d_k}$ are the key and value, respectively.

Causal self-attention. We also use causal self-attention [28] in T2M-GPT+ for autoregressive modeling. The attention can be computed as:

$$CSA = \text{Softmax} \left(\frac{Q(m)K(m)^\top}{\sqrt{d_k}} + \text{mask} \right) \cdot V(m), \quad (6)$$

where $K(m) \in \mathbb{R}^{\frac{T}{l} \times d_k}$ and $V(m) \in \mathbb{R}^{\frac{T}{l} \times d_k}$ are the key and value respectively. The causal mask $\text{mask}_{i,j} = -\infty \times \mathbf{1}(i < j) + 0 \times \mathbf{1}(i \geq j)$ is used to enforce the restriction that future information cannot affect the computation of the current tokens, where $\mathbf{1}(\cdot)$ is the indicator function. For inference, we generate code indices in an autoregressive manner. The generation process begins with the [START] token and stops when the model predicts the [END] token. Our approach enables sampling from predicted probability distributions to synthesize diverse motions.

Corrupted sequences for the training-testing discrepancy. During training, the model predicts the i -th index based on $i-1$ ground truth indices. However, there is no guarantee that the previously generated indices are accurate during inference. This discrepancy leads to performance degradation. To address this issue, we introduce a straightforward data augmentation strategy that simulates potential errors. Specifically, during training, we replace $\tau \times 100\%$ of the ground-truth code indices with incorrect indices randomly selected from the codebook. τ can either be a fixed hyperparameter or dynamically sampled from a uniform distribution $\tau \in \mathcal{U}[0, 1]$. We present ablative experiments in Section IV-D to demonstrate the effectiveness of this strategy.

C. T2M-GIT+

T2M-GPT+ generates motion sequences with the next-index prediction based on previously generated tokens, which is time-consuming and error-sensitive to previous results. We explore a more efficient method, T2M-GIT+ (see Figure 2 (c)), which leverages parallel decoding. It introduces the Masked Token Modeling (MTM) [29] strategy to generate index tokens in parallel. Specifically, with a sequence of indices $S = [s_1, s_2, \dots, s_{T/l}]$ available, a percentage of

$\gamma \times 100\%$ tokens are randomly sampled and replaced by the [MASK] token leading to $\tilde{S} = [\tilde{s}_1, \tilde{s}_2, \dots, \tilde{s}_{T/l}]$ during training, where $\gamma \in (0, 1]$ and $\tilde{s}_i \in [1, 2, \dots, K, [\text{MASK}]]$. Then, we formulate the model training as a non-autoregressive masked code-index prediction task: given masked index tokens $\tilde{S}$ and the condition c , we aim to predict the distribution $p(S|c, \tilde{S})$.

Optimization goal. Similar to T2M-GPT+, we denote the likelihood of the sequence as $p(S|c, \tilde{S}) = \prod_{i=1}^{|\tilde{S}|} p(s_i|c, \tilde{S})$, and maximize the log-likelihood of the distribution:

$$\mathcal{L}_{GIT} = \mathbb{E}_{S \sim p(S)} [-\log p(S|c, \tilde{S})]. \quad (7)$$

When generating the corresponding motion index, we follow Section III-B to use LLM’s word-wise embeddings and employ cross-attention to attend to the text embedding. Different from T2M-GPT+, we use a standard self-attention operator rather than causal self-attention.

Inference. In terms of inference, we apply iterative decoding [29] to generate motion index sequences. For each iteration, we sample the $\frac{T}{l} - M$ most confident tokens based on the predicted probability of each token, and then mask the remaining tokens for the next iteration. M can be calculated according to the scheduling function $f(\cdot)$ (e.g., cosine decay) with $\lceil f(\frac{N}{N}) \times \frac{T}{l} \rceil$, where N is the total number of iteration steps. Therefore, a motion sequence can be generated in N steps where $N < \frac{T}{l}$. We provide an ablation study of the number of iterations N in Section IV-D.

IV. EXPERIMENTS

In this section, we provide the results of our experiments. First, we introduce benchmarks and evaluation metrics in Section IV-A. Next, we present implementation details in Section IV-B. Finally, we compare our approach with state-of-the-art methods in Section IV-C and provide an in-depth analysis and discussion in Section IV-D.

A. Datasets and evaluation metrics

We evaluate our approach on two widely-used benchmark datasets: HumanML3D [7] and KIT Motion-Language (KIT-ML) [32]. We use the evaluation metrics proposed in [7].

HumanML3D. HumanML3D [7] is a large-scale benchmark for human motion generation from texts. It comprises 14,616 motion sequences and 44,970 text annotations. Following Guo *et al.* [7], the dataset is divided into training, validation, and test sets with 80%, 5%, and 15% of the data, respectively.

KIT Motion-Language (KIT-ML). KIT-ML [32] is another leading dataset for text-to-motion generation. It contains 3,911 motion sequences and 6,278 texts. The dataset is split into training, validation, and test sets with 4888/300/830 samples, respectively.

Evaluation metrics. Following previous works [7]–[10], [16], we first extract global representations of the motion and text through an evaluation model from Guo *et al.* [7]. The global representations are used to measure the performance of text-driven motion generation using the following metrics: R-Precision (Top-1/2/3 motion-text retrieval accuracy), Frechet

TABLE I

QUANTITATIVE COMPARISON ON THE HUMANML3D [7] TEST SET. WE REPEAT THE EVALUATION 20 TIMES, THEN REPORT THE AVERAGE METRIC ALONG WITH A 95% CONFIDENCE INTERVAL. THE BEST RESULT AND THE SECOND BEST RESULT ARE IN BOLD AND UNDERLINED. THE § INDICATES METHODS OFFICIALLY PUBLISHED AFTER THE INITIAL SUBMISSION OF THIS PAPER.

Methods	R-Precision $\uparrow$			FID $\downarrow$	MM-Dist $\downarrow$	Diversity $\uparrow$	MModality $\uparrow$
	Top-1	Top-2	Top-3				
Real motion	0.511 $\pm$.003	0.703 $\pm$.003	0.797 $\pm$.002	0.002 $\pm$.000	2.974 $\pm$.008	9.503 $\pm$.065	-
Our VQ-VAE (Recons.)	0.501 $\pm$.002	0.692 $\pm$.002	0.785 $\pm$.002	0.070 $\pm$.001	3.072 $\pm$.009	9.593 $\pm$.079	-
TEMOS [5]	0.424 $\pm$.002	0.612 $\pm$.002	0.722 $\pm$.002	3.734 $\pm$.028	3.703 $\pm$.008	8.973 $\pm$.071	0.368 $\pm$.018
TM2T [8]	0.424 $\pm$.003	0.618 $\pm$.003	0.729 $\pm$.002	1.501 $\pm$.017	3.467 $\pm$.011	8.589 $\pm$.076	2.424 $\pm$.093
Guo <i>et al.</i> [7]	0.455 $\pm$.003	0.636 $\pm$.003	0.736 $\pm$.002	1.087 $\pm$.021	3.347 $\pm$.008	9.175 $\pm$.083	2.219 $\pm$.074
MLD [16]	0.481 $\pm$.003	0.673 $\pm$.003	0.772 $\pm$.002	0.473 $\pm$.013	3.196 $\pm$.010	9.724 $\pm$.082	2.413 $\pm$.079
MDM [14]	-	-	0.611 $\pm$.007	0.544 $\pm$.044	5.566 $\pm$.027	9.559 $\pm$.086	2.799 $\pm$.072
MotionDiffuse [10]	0.491 $\pm$.001	0.681 $\pm$.001	0.782 $\pm$.001	0.630 $\pm$.001	3.113 $\pm$.001	9.410 $\pm$.049	1.553 $\pm$.042
Jiang <i>et al.</i> [93]	0.492 $\pm$.003	0.681 $\pm$.003	0.778 $\pm$.002	0.232 $\pm$.008	3.096 $\pm$.008	9.528 $\pm$.071	2.008 $\pm$.084
ReMoDiffuse [79]§	0.510 $\pm$.005	0.698 $\pm$.006	0.795 $\pm$.004	0.103 $\pm$.004	2.974 $\pm$.016	9.018 $\pm$.075	1.795 $\pm$.043
AttT2M [74]§	0.499 $\pm$.003	0.690 $\pm$.002	0.786 $\pm$.002	0.112 $\pm$.006	3.038 $\pm$.007	9.700 $\pm$.090	2.452 $\pm$.051
MoMask [78]§	0.521 $\pm$.002	0.713 $\pm$.002	0.807 $\pm$.002	0.045 $\pm$.002	2.958 $\pm$.008	-	1.241 $\pm$.040
MMM [77]§	0.504 $\pm$.003	0.696 $\pm$.003	0.794 $\pm$.002	0.080 $\pm$.003	2.998 $\pm$.007	9.411 $\pm$.058	1.164 $\pm$.041
MotionLCM [81]§	0.502 $\pm$.003	0.698 $\pm$.002	0.798 $\pm$.002	0.304 $\pm$.012	3.012 $\pm$.007	9.607 $\pm$.066	2.259 $\pm$.092
LMM [82]§	0.525 $\pm$.002	0.719 $\pm$.002	0.811 $\pm$.002	0.040 $\pm$.002	2.943 $\pm$.012	9.814 $\pm$.076	2.683 $\pm$.054
T2M-GPT (CLIP) [9]	0.492 $\pm$.003	0.679 $\pm$.002	0.775 $\pm$.002	0.141 $\pm$.005	3.121 $\pm$.009	9.722 $\pm$.082	1.831 $\pm$.048
T2M-GPT+ (DistilBERT)	0.530 $\pm$.005	<u>0.721</u> $\pm$.005	0.812 $\pm$.005	0.115 $\pm$.011	2.883 $\pm$.017	9.574 $\pm$.042	1.782 $\pm$.085
T2M-GPT+ (Llama-7B)	0.531 $\pm$.004	<u>0.721</u> $\pm$.002	0.812 $\pm$.003	<u>0.103</u> $\pm$.008	2.885 $\pm$.018	9.594 $\pm$.071	1.652 $\pm$.004
T2M-GPT+ (Llama-13B)	<u>0.532</u> $\pm$.006	0.720 $\pm$.004	<u>0.814</u> $\pm$.004	0.099 $\pm$.008	2.898 $\pm$.017	9.676 $\pm$.102	1.812 $\pm$.051
T2M-GIT+ (DistilBERT)	0.526 $\pm$.004	0.718 $\pm$.002	0.812 $\pm$.004	0.164 $\pm$.010	<u>2.865</u> $\pm$.019	9.658 $\pm$.115	0.378 $\pm$.03
T2M-GIT+ (Llama-7B)	0.527 $\pm$.001	0.719 $\pm$.001	<u>0.814</u> $\pm$.003	0.135 $\pm$.008	2.881 $\pm$.015	9.748 $\pm$.022	0.452 $\pm$.015
T2M-GIT+ (Llama-13B)	0.535 $\pm$.004	0.729 $\pm$.004	0.821 $\pm$.003	0.129 $\pm$.008	2.849 $\pm$.010	<u>9.787</u> $\pm$.022	0.404 $\pm$.03

Inception Distance (FID), Multimodal Distance (MM-Dist), Diversity, and Multimodality (MModality). To ensure robustness, we conduct the evaluation for each metric 20 times and report the average along with the 95% confidence interval.

B. Implementation details

For Motion VQ-VAE, we crop motion sequences to a fixed length $T = 64$ on the two datasets (*i.e.*, HumanML3D and KIT-ML) during training. Following Guo *et al.* [7], the dimensions of motion features are 263 and 251 for HumanML3D and KIT-ML, which consist of local joint position, velocity, root rotation, global translation, and global rotation. These features are calculated from 22 and 21 SMPL [94] joints, respectively. The codebook includes 512 codes with 512 dimensions. The optimization process leverages the AdamW optimizer [95] with $[\beta_1, \beta_2] = [0.9, 0.99]$. Training is conducted over 300K iterations, with an initial learning rate of 2×10^{-4} for the first 200K iterations, followed by 1×10^{-5} for the remaining 100K. The batch size is set to 256, the exponential moving average constant $\lambda = 0.99$, and the downsampling rate $l = 4$. We set β and α to 0.02 and 0.5 for $\mathcal{L}_{vq}$ and $\mathcal{L}_{re}$, respectively.

For the T2M-GPT+ and T2M-GIT+, we use the architecture comprising 18 transformer layers [27] with a dimension of 1024. We conduct experiments with Llama-13B and Llama-7B [33] models as the text feature extractor. To improve efficiency, we run the LLM once for each textual description to extract text embeddings, which are then cached for subsequent training and evaluation. We train two extra [START] and [END] tokens as signals to start and stop index generation in T2M-GPT+. The optimization process leverages the AdamW

optimizer [95] with $[\beta_1, \beta_2] = [0.5, 0.99]$. We train the transformer for 300K iterations in total with a batch size of 256. The learning rate is initialized as 2×10^{-4} and decayed to 1×10^{-4} after 150K iterations. All our experiments are conducted on a single NVIDIA A100 GPU. Training Motion VQ-VAE and transformers (T2M-GPT+ and T2M-GIT+) takes about 12 hours and 3.8 days, respectively. We provide inference time in Table III and Section IV-D. Moreover, we use cosine and square mask scheduling functions in T2M-GIT+ for HumanML3D and KIT-ML, respectively.

C. Comparison to state-of-the-art approaches

Quantitative comparison. We show the quantitative results in Table I and Table II on the HumanML3D [7] and KIT-ML [32] test sets. Our VQ-VAE-based reconstruction achieves performance close to that of real motion on both datasets. These results demonstrate the effectiveness of our VQ-VAE in learning high-quality discrete motion representations. For the generation, our approach significantly outperforms the previous state-of-the-art method, *i.e.*, MotionDiffuse [10]. Compared to T2M-GPT [9], simply using LLMs as the text encoder improves the performance on text-motion consistency (R-Precision and MM-Dist) a lot. T2M-GIT+ achieves comparable performance to T2M-GPT+ but with much faster decoding. A more detailed analysis of inference speed is provided in Section IV-D. During the review period, several new methods were published. We therefore further compare our approach with more recent works. Despite being proposed three years ago, our model remains competitive with methods officially published after our initial submission. In particular,

TABLE II

QUANTITATIVE COMPARISON ON THE KIT-ML [32] TEST SET. WE REPEAT THE EVALUATION 20 TIMES, THEN REPORT THE AVERAGE METRIC ALONG WITH A 95% CONFIDENCE INTERVAL. THE BEST RESULT AND THE SECOND BEST RESULT ARE IN BOLD AND UNDERLINED. THE § INDICATES METHODS OFFICIALLY PUBLISHED AFTER THE INITIAL SUBMISSION OF THIS PAPER.

Methods	R-Precision $\uparrow$			FID $\downarrow$	MM-Dist $\downarrow$	Diversity $\uparrow$	MModality $\uparrow$
	Top-1	Top-2	Top-3				
Real motion	0.424 $\pm$.005	0.649 $\pm$.006	0.779 $\pm$.006	0.031 $\pm$.004	2.788 $\pm$.012	11.08 $\pm$.097	-
Our VQ-VAE (Recons.)	0.399 $\pm$.005	0.614 $\pm$.005	0.740 $\pm$.006	0.472 $\pm$.011	2.986 $\pm$.027	10.994 $\pm$.120	-
TEMOS [5]	0.353 $\pm$.002	0.561 $\pm$.002	0.687 $\pm$.002	3.717 $\pm$.028	3.417 $\pm$.008	10.84 $\pm$.071	0.532 $\pm$.018
TM2T [8]	0.280 $\pm$.006	0.463 $\pm$.007	0.587 $\pm$.005	3.599 $\pm$.051	4.591 $\pm$.019	9.473 $\pm$.100	3.292 $\pm$.034
Guo <i>et al.</i> [7]	0.361 $\pm$.006	0.559 $\pm$.007	0.681 $\pm$.007	3.022 $\pm$.107	3.488 $\pm$.028	10.72 $\pm$.145	2.052 $\pm$.107
MLD [16]	0.390 $\pm$.008	0.609 $\pm$.008	0.734 $\pm$.007	0.404 $\pm$.027	3.204 $\pm$.027	10.80 $\pm$.117	2.192 $\pm$.071
MDM [14]	-	-	0.396 $\pm$.004	0.497 $\pm$.021	9.191 $\pm$.022	10.847 $\pm$.109	1.907 $\pm$.214
MotionDiffuse [10]	0.417 $\pm$.004	0.621 $\pm$.004	0.739 $\pm$.004	1.954 $\pm$.062	2.958 $\pm$.005	<u>11.10</u> $\pm$.143	0.730 $\pm$.013
ReMoDiffuse [79]§	0.427 $\pm$.014	0.641 $\pm$.004	0.765 $\pm$.055	<u>0.155</u> $\pm$.006	2.814 $\pm$.012	10.800 $\pm$.105	1.239 $\pm$.028
AttT2M [74]§	0.413 $\pm$.006	0.632 $\pm$.006	0.751 $\pm$.006	0.870 $\pm$.039	3.039 $\pm$.021	10.96 $\pm$.123	<u>2.281</u> $\pm$.047
MoMask [78]§	0.433 $\pm$.007	0.656 $\pm$.005	0.781 $\pm$.005	0.204 $\pm$.011	2.779 $\pm$.022	-	1.131 $\pm$.043
MMM [77]§	0.381 $\pm$.005	0.590 $\pm$.006	0.718 $\pm$.005	0.429 $\pm$.019	3.146 $\pm$.019	10.633 $\pm$.097	1.105 $\pm$.026
LMM [82]§	0.430 $\pm$.015	0.653 $\pm$.017	0.779 $\pm$.014	0.137 $\pm$.023	<u>2.791</u> $\pm$.018	11.24 $\pm$.103	1.885 $\pm$.127
T2M-GPT (CLIP) [9]	0.416 $\pm$.006	0.627 $\pm$.006	0.745 $\pm$.006	0.514 $\pm$.029	3.007 $\pm$.023	10.921 $\pm$.108	1.570 $\pm$.039
T2M-GPT+ (Llama-13B)	0.412 $\pm$.004	0.629 $\pm$.007	0.755 $\pm$.006	0.504 $\pm$.042	2.944 $\pm$.019	10.947 $\pm$.120	1.685 $\pm$.042
T2M-GIT+ (Llama-13B)	0.422 $\pm$.007	0.632 $\pm$.006	0.751 $\pm$.006	0.524 $\pm$.030	2.945 $\pm$.023	10.952 $\pm$.096	0.702 $\pm$.015

TABLE III

IMPACT OF ITERATION NUMBER AND INFERENCE SPEED ON HUMANML3D [7]. WE REPORT R-PRECISION, FID, MM-DIST, AND AVERAGE INFERENCE TIME PER SENTENCE (IN SECONDS) ACROSS ALL SETTINGS, USING THE SAME TEXT ENCODER (LLAMA-13B). THE BEST AND SECOND-BEST RESULTS ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY. VALUES IN PARENTHESES DENOTE THE SPEEDUP RELATIVE TO T2M-GPT+

Methods	Timesteps	Inference time (s)	R-Precision $\uparrow$			FID $\downarrow$	MM-Dist $\downarrow$
			Top-1	Top-2	Top-3		
T2M-GIT+	$N = 6$	0.049 ($\times 6.20\uparrow$)	0.527 $\pm$.003	0.719 $\pm$.003	0.812 $\pm$.002	0.137 $\pm$.006	2.878 $\pm$.007
	$N = 10$	0.077 ($\times 3.95\uparrow$)	0.530 $\pm$.003	<u>0.722</u> $\pm$.003	<u>0.818</u> $\pm$.002	0.136 $\pm$.004	<u>2.862</u> $\pm$.009
	$N = 14$	0.115 ($\times 2.64\uparrow$)	0.535 $\pm$.004	0.729 $\pm$.004	0.821 $\pm$.003	<u>0.129</u> $\pm$.008	2.849 $\pm$.010
	$N = 18$	0.138 ($\times 2.20\uparrow$)	0.526 $\pm$.002	0.719 $\pm$.003	0.814 $\pm$.002	0.134 $\pm$.005	2.874 $\pm$.009
T2M-GPT+	T/l	0.304	<u>0.532</u> $\pm$.006	0.720 $\pm$.004	0.814 $\pm$.004	0.099 $\pm$.008	2.898 $\pm$.017

TABLE IV

USER STUDIES ON HUMANML3D [7]. IN USER STUDY 1, WE COMPARE OUR METHOD WITH SEVERAL STATE-OF-THE-ART METHODS, *i.e.*, MDM [14], MLD [16], MOTIONDIFFUSE [10], AND T2M-GPT [9]. IN USER STUDY 2, WE COMPARE T2M-GPT+ AND T2M-GIT+. THE VALUES IN THE TABLE REPRESENT THE PERCENTAGE OF RATERS SELECTING WHICH METHOD THEY DEEM BETTER.

Methods	User study 1 (%)		User study 2 (%)
	T2M-GPT+	T2M-GIT+	
MDM [14]	5.23	8.86	-
MLD [16]	4.62	3.43	-
MotionDiffuse [10]	4.62	4.00	-
T2M-GPT [9]	17.23	20.00	-
T2M-GPT+	68.31	-	39.27
T2M-GIT+	-	63.71	60.73

on HumanML3D, our approach achieves better text-motion alignment while maintaining a comparable FID. Note that we implicitly learn the motion length through an additional [END] token for T2M-GPT+, and evaluate the T2M-GIT+ model with the ground-truth motion length.

Furthermore, we conduct two human evaluations on the HumanML3D dataset in Table IV. The user study is conducted

TABLE V

IMPACT OF QUANTIZATION STRATEGIES ON THE HUMANML3D [7] TEST SET. WE UTILIZE THE SAME VQ-VAE ARCHITECTURE IN SECTION III-A FOR CODE RESET AND EMA. WE PROVIDE FID AND TOP-1 ACCURACY TO EVALUATE THE PERFORMANCE OF MOTION RECONSTRUCTION.

Quantizer		Reconstruction	
Code Reset	EMA	FID $\downarrow$	Top-1 $\uparrow$
✓	✓	0.492 $\pm$.004	0.436 $\pm$.003
		0.097 $\pm$.001	0.499 $\pm$.002
		0.102 $\pm$.001	0.494 $\pm$.003
		0.070 $\pm$.001	0.501 $\pm$.002

across 25 human motions. We recruited 14 participants for the user studies. We asked users to rate based on two criteria: (1) motion quality, (2) text-motion consistency. In User Study 1, we compare the proposed methods with several state-of-the-art methods, *i.e.*, MDM [14], MLD [16], MotionDiffuse [10], and T2M-GPT [9]. It can be seen that the proposed T2M-GPT+ and T2M-GIT+ are both preferred by users. We also present a comparison between T2M-GPT+ and T2M-GIT+ in User Study 2, and T2M-GIT+ receives better feedback.

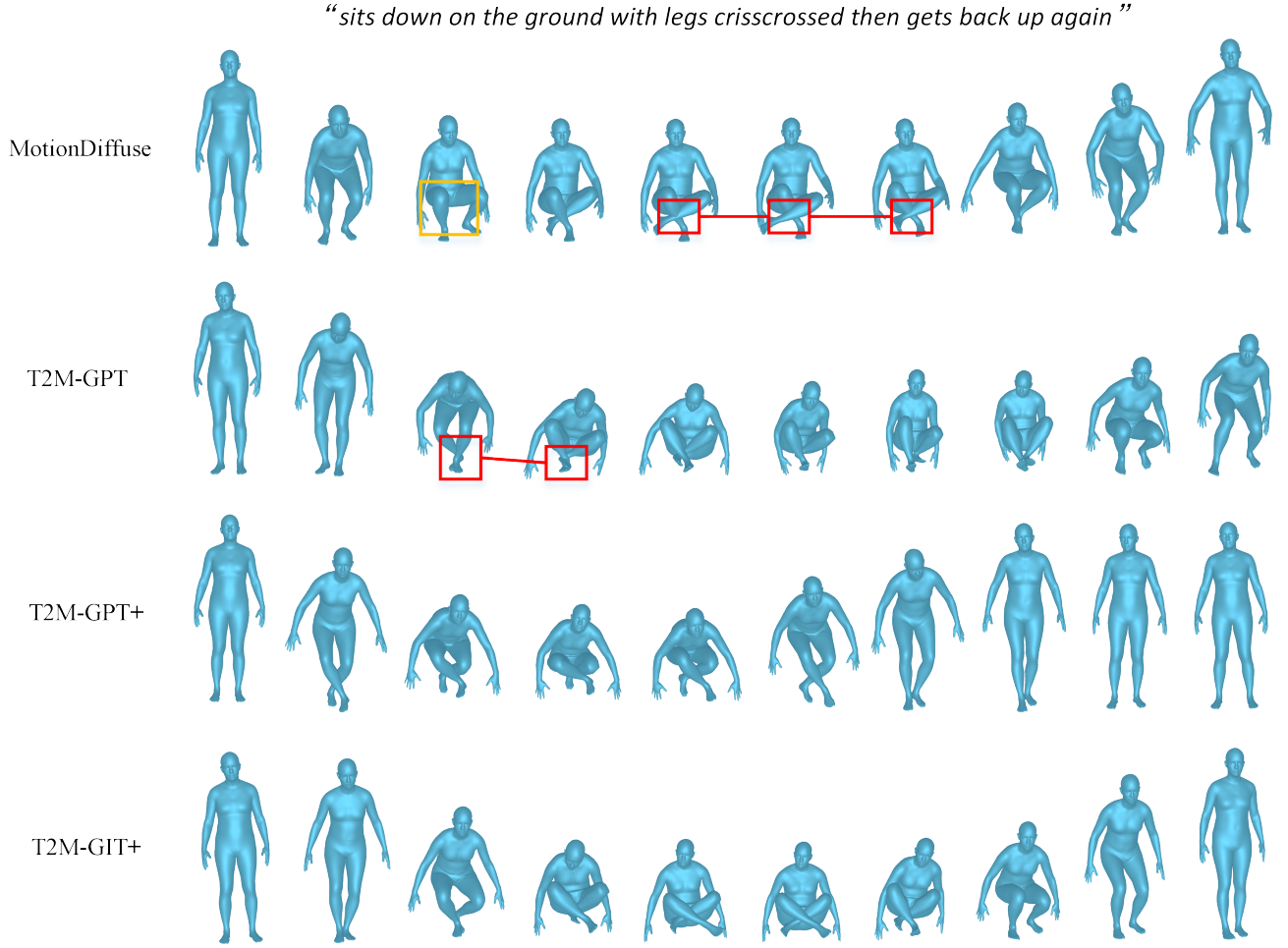


Fig. 4. Visual results on the HumanML3D [7] dataset. We compare our generation results with MotionDiffuse [10], and T2M-GPT [9]. Distorted motions (red) and the overhang problem (yellow) are highlighted.

TABLE VI
ANALYSIS OF τ ON THE HUMANML3D [7] TEST SET.

τ	FID ↓	Top-1 ↑	MM-Dist ↓
0.0	0.099 $\pm$.010	0.414 $\pm$.005	3.753 $\pm$.018
0.3	0.088 $\pm$.009	0.517 $\pm$.004	2.967 $\pm$.010
0.5	0.099 $\pm$.008	0.532 $\pm$.006	2.898 $\pm$.017
0.7	0.203 $\pm$.012	0.525 $\pm$.003	2.929 $\pm$.012
0.9	1.171 $\pm$.053	0.465 $\pm$.003	3.354 $\pm$.012
$\mathcal{U}[0, 1]$	0.161 $\pm$.010	0.532 $\pm$.004	2.876 $\pm$.011

TABLE VII
ANALYSIS OF τ ON THE KIT-ML [32] TEST SET.

τ	FID ↓	Top-1 ↑	MM-Dist ↓
0.0	0.624 $\pm$.056	0.396 $\pm$.005	3.205 $\pm$.027
0.3	0.681 $\pm$.029	0.409 $\pm$.007	3.117 $\pm$.021
0.5	0.504 $\pm$.042	0.412 $\pm$.004	2.944 $\pm$.019
0.7	0.712 $\pm$.032	0.406 $\pm$.006	3.118 $\pm$.027
0.9	0.784 $\pm$.057	0.384 $\pm$.006	3.353 $\pm$.026
$\mathcal{U}[0, 1]$	0.667 $\pm$.030	0.403 $\pm$.006	3.100 $\pm$.031

TABLE VIII
ANALYSIS OF $f(\cdot)$ ON THE HUMANML3D [7] TEST SET.

$f(\cdot)$	FID ↓	Top-1 ↑	MM-Dist ↓
cosine	0.129 $\pm$.008	0.535 $\pm$.004	2.849 $\pm$.010
linear	0.127 $\pm$.005	0.523 $\pm$.003	2.893 $\pm$.008
square	0.124 $\pm$.004	0.528 $\pm$.002	2.874 $\pm$.008

TABLE IX
ANALYSIS OF $f(\cdot)$ ON THE KIT-ML [32] TEST SET.

$f(\cdot)$	FID ↓	Top-1 ↑	MM-Dist ↓
cosine	0.571 $\pm$.040	0.417 $\pm$.007	2.932 $\pm$.026
linear	0.641 $\pm$.035	0.407 $\pm$.007	2.991 $\pm$.021
square	0.524 $\pm$.030	0.422 $\pm$.007	2.945 $\pm$.023

than competitive approaches. We highlight unrealistic motions (one leg passes through another leg) in red that are generated by MotionDiffuse [10] and T2M-GPT [9]. Some consecutive motion frames generated by MotionDiffuse [10] are suspended in the air (highlighted in yellow).

Qualitative comparison. We provide visual results and compare our approach with the state-of-the-art models: MotionDiffuse [10] and T2M-GPT [9] in Figure 4. We find that T2M-GPT+ and T2M-GIT+ generate motions with better quality

D. Analysis and discussion

Analysis of iteration number N and inference speed. We study the iteration number N as well as the inference time of



Fig. 5. Visual results of T2M-GIT+ generation process on HumanML3D [7]. We compare the visual results of different generation steps of T2M-GIT+.

both T2M-GIT+ and T2M-GPT+. As shown in Table III, $N = 14$ achieves the best performance, and fewer iterations yield similar performance. In terms of inference time, it can be seen that T2M-GIT+ is significantly more efficient. For example, T2M-GIT+ with $N = 14$ is about 2.64 times faster than T2M-GPT+. With a batch size of 1, T2M-GPT+ and T2M-GIT+ take about 22 minutes and 8.5 minutes for a single evaluation on the HumanML3D [7] test set. Visual results of different N are provided in Figure 5. We observe that in the first few steps, the motion is unrealistic (especially the walking part), and then after some iterations, T2M-GIT+ generates motion

with better quality. It is worth noting that the correspondence between motion and discrete representations is not strictly one-to-one. The representation of a human motion is influenced by the motion token itself and the arrangement of its surrounding tokens, leading to variations in motion across different time steps.

Impact of dataset size. We analyze the impact of dataset size, and the results are shown in Figure 6. Our goal is to explore whether the data size of HumanML3D [7] is sufficient for the text-to-motion generation task. We conduct experiments by training our models (VQ-VAE, T2M-GPT+, and T2M-GIT+)

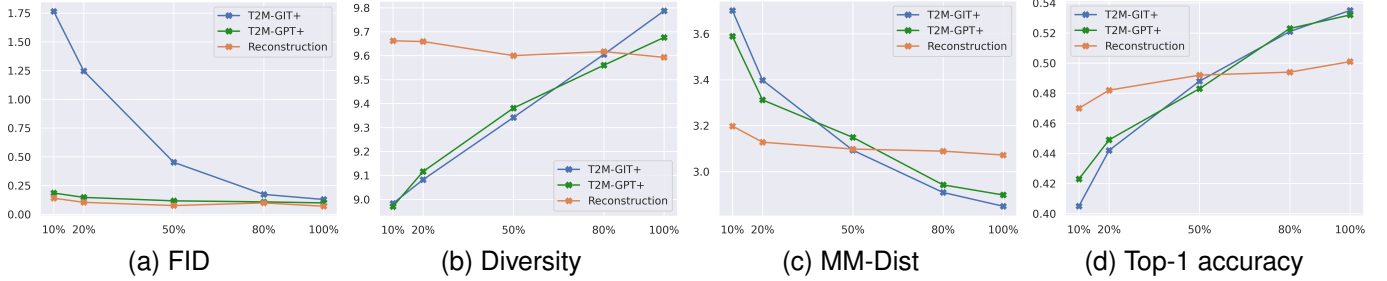


Fig. 6. Impact of dataset size on HumanML3D [7]. We train our models, *i.e.*, VQ-VAE (*Reconstruction*), T2M-GIT+, and T2M-GPT+, on the subsets of HumanML3D [7] under 10%, 20%, 50%, 80%, and 100% partitions. We evaluate all the models on the test set and report FID, Diversity, MM-Dist, and Top-1 accuracy. The results suggest that our approach might be further improved with more training data.

TABLE X
ABLATION OF VQ-VAE LOSS FUNCTIONS ON HUMANML3D [7].

$\mathcal{L}_{re}$	α	Reconstruction	
		FID ↓	Top-1
L1	0	$0.095 \pm .001$	$0.493 \pm .002$
L1	0.5	$0.144 \pm .001$	$0.495 \pm .003$
L1	1	$0.160 \pm .001$	$0.496 \pm .003$
L1-Smooth	0	$0.112 \pm .001$	$0.496 \pm .003$
L1-Smooth	0.5	$0.070 \pm .001$	$0.501 \pm .002$
L1-Smooth	1	$0.128 \pm .001$	$0.499 \pm .003$
L2	0	$0.321 \pm .002$	$0.478 \pm .003$
L2	0.5	$0.292 \pm .002$	$0.483 \pm .002$
L2	1	$0.213 \pm .002$	$0.490 \pm .003$

TABLE XI
STUDY OF THE CODEBOOK SIZE ON HUMANML3D [7].

Num. code	Reconstruction	
	FID ↓	Top-1
256	$0.145 \pm .001$	$0.497 \pm .002$
512	$0.070 \pm .001$	$0.501 \pm .002$
1024	$0.090 \pm .001$	$0.498 \pm .003$

on varying proportions of the training set (10%, 20%, 50%, 80%, and 100%), and evaluate the models on the entire test set. Our key insights are two-fold: *i)* T2M-GPT+ generates good-quality motions (FID) when trained on 10% data, but the motions lack consistency with textual descriptions (MM-Dist and Top-1 accuracy). This indicates that both motion quality (FID) and motion-text consistency (MM-Dist and Top-1 accuracy) should be considered simultaneously during evaluation; *ii)* For T2M-GPT+ and T2M-GIT+, the performance consistently improves with increased training data, suggesting that our approach could be further improved with more training data.

Quantization strategies. We investigate the impact of different quantization strategies, and the results are shown in Table V. We observe that the motions reconstructed by naive VQ-VAE are not consistent with texts. With *EMA* or *Code Reset*, our approach significantly improves the naive VQ-VAE baseline on FID and Top-1 accuracy, which demonstrates the effectiveness of these two strategies.

Study on τ for the corruption strategy. We explore the impact of τ , which is the corruption ratio used for T2M-GPT+ in Section III-B. We conduct ablative experiments on the HumanML3D and KIT-ML datasets. The results are illustrated in Table VI and Table VII. It can be seen that, with $\tau = 0.3, 0.5, 0.7$ and $\tau \in \mathcal{U}[0, 1]$, Top-1 accuracy is significantly improved over $\tau = 0.0$. On HumanML3D [7], $\tau = 0.5$ achieves comparable Top-1 accuracy and MM-Dist results to $\tau \in \mathcal{U}[0, 1]$, while outperforming it in terms of FID. On KIT-ML [32], $\tau = 0.5$ yields the best results.

Ablation on scheduling function $f(\cdot)$ in T2M-GIT+. We conduct additional ablative experiments on the mask scheduling functions to further investigate T2M-GIT+. We compare three commonly used functions: cosine, linear, and square. The results are shown in Table VIII and Table IX. We observe that the cosine function is preferred on HumanML3D [7], and the square function performs better on the KIT-ML [32] dataset.

Impact of the reconstruction loss in motion VQ-VAE. We study the reconstruction loss ($\mathcal{L}_{re}$ in Equation 3) and the weight α of the velocity loss (Equation 3). The results are shown in Table X. We find that Smooth L1 loss outperforms L1 and L2 losses. Additionally, $\alpha = 0.5$ leads to the best performance.

Ablation study of the number of codes in VQ-VAE. As illustrated in Table XI, increasing the codebook size from 256 to 512 significantly improves the performance. Although using 1024 codes still achieves competitive results, 512 codes are sufficient for motion reconstruction.

V. CONCLUSION

In this work, we investigate a classic framework based on VQ-VAE and Transformers (T2M-GPT+ and T2M-GIT+) for synthesizing human motion from textual descriptions. Our method significantly outperforms diffusion-based approaches and T2M-GPT. We demonstrate that word embeddings from an autoregressive large language model can substantially improve performance without specific fine-tuning. We also investigate parallel decoding and show that T2M-GIT+ largely improves inference speed while achieving performance comparable to T2M-GPT+. We conduct a detailed analysis of various quantization strategies. Moreover, we explore the impact of dataset

size. Our results indicate that a larger dataset can further enhance the performance of our approach.

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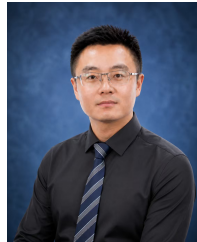
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